

NTA JEE Mains Jan 2026

Test Date	23/01/2026
Test Time	9:00 AM - 12:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1 Let the domain of the function $f(x) = \log_3 \log_5 \log_7 (9x - x^2 - 13)$ be the interval (m, n) . Let the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ have eccentricity $\frac{n}{3}$ and the length of the latus rectum $\frac{8m}{3}$. Then $b^2 - a^2$ is equal to :

Options

1. 9
2. 7
3. 5
4. 11

Question Type : MCQ

Question ID : 8606541360

Option 1 ID : 8606544629

Option 2 ID : 8606544628

Option 3 ID : 8606544627

Option 4 ID : 8606544630

Q.2 Let the mean and variance of 8 numbers $-10, -7, -1, x, y, 9, 2, 16$ be $\frac{7}{2}$ and $\frac{293}{4}$, respectively.

Then the mean of 4 numbers $x, y, x + y + 1, |x - y|$ is :

Options

1. 12
2. 10
3. 11
4. 9

Question Type : MCQ

Question ID : 8606541358

Option 1 ID : 8606544622

Option 2 ID : 8606544620

Option 3 ID : 8606544621

Option 4 ID : 8606544619

Q.3

The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are (1, 6, 3) and (4, 9, α) respectively and C is at a distance of 10 units from B. The area (in sq. units) of ΔABC is :

Options

1. $5\sqrt{13}$
2. $15\sqrt{13}$
3. $20\sqrt{13}$
4. $10\sqrt{13}$

Question Type : **MCQ**

Question ID : **8606541364**

Option 1 ID : **8606544644**

Option 2 ID : **8606544646**

Option 3 ID : **8606544645**

Option 4 ID : **8606544643**

Q.4

The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt[3]{\tan 2x}}$ is :

Options

1. $\frac{\pi}{12}$
2. $\frac{\pi}{6}$
3. $\frac{\pi}{3}$
4. $\frac{\pi}{18}$

Question Type : **MCQ**

Question ID : **8606541369**

Option 1 ID : **8606544665**

Option 2 ID : **8606544664**

Option 3 ID : **8606544663**

Option 4 ID : **8606544666**

Q.5

Let $\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} - \hat{j} - 3\hat{k}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$.

Then $(\vec{a} - \vec{b}) \cdot \vec{d}$ is equal to :

Options

1. 2
2. 4
3. -4
4. -2

Question Type : MCQ

Question ID : 8606541365

Option 1 ID : 8606544649

Option 2 ID : 8606544647

Option 3 ID : 8606544648

Option 4 ID : 8606544650

Q.6

Let $y = y(x)$ be the solution of the differential equation $x^4 dy + (4x^3 y + 2 \sin x) dx = 0$, $x > 0$, $y\left(\frac{\pi}{2}\right) = 0$.

Then $\pi^4 y\left(\frac{\pi}{3}\right)$ is equal to :

Options

1. 81
2. 64
3. 92
4. 72

Question Type : MCQ

Question ID : 8606541370

Option 1 ID : 8606544667

Option 2 ID : 8606544669

Option 3 ID : 8606544670

Option 4 ID : 8606544668

Q.7

Let the line $y - x = 1$ intersect the ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$ at the points A and B. Then the angle made by the line segment AB at the center of the ellipse is :

Options

1. $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$
2. $\pi - \tan^{-1}\left(\frac{1}{4}\right)$
3. $\frac{\pi}{2} + 2 \tan^{-1}\left(\frac{1}{4}\right)$
4. $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$

Question Type : MCQ

Question ID : 8606541362

Option 1 ID : 8606544636

Option 2 ID : 8606544635

Option 3 ID : 8606544638

Option 4 ID : 8606544637

Q.8

$$\text{Let } f(x) = \begin{cases} \frac{ax^2 + 2ax + 3}{4x^2 + 4x - 3} & , x \neq -\frac{3}{2}, \frac{1}{2} \\ b & , x = -\frac{3}{2}, \frac{1}{2} \end{cases}$$

be continuous at $x = -\frac{3}{2}$. If $f \circ f(x) = \frac{7}{5}$, then x is equal to :

Options

1. 1
2. 2
3. 0
4. 1.4

Question Type : MCQ

Question ID : 8606541367

Option 1 ID : 8606544656

Option 2 ID : 8606544658

Option 3 ID : 8606544655

Option 4 ID : 8606544657

Q.9 The sum of all possible values of $n \in \mathbb{N}$, so that the coefficients of x , x^2 and x^3 in the expansion of $(1+x^2)^2(1+x)^n$, are in arithmetic progression is :

Options

1. 9
2. 12
3. 7
4. 3

Question Type : MCQ

Question ID : 8606541356

Option 1 ID : 8606544613

Option 2 ID : 8606544614

Option 3 ID : 8606544612

Option 4 ID : 8606544611

Q.10 Let $S = \{z : 3 \leq |2z - 3(1+i)| \leq 7\}$ be a set of complex numbers.

Then $\min_{z \in S} \left| z + \frac{1}{2}(5+3i) \right|$ is equal to :

Options

1. 2
2. $\frac{5}{2}$
3. $\frac{1}{2}$
4. $\frac{3}{2}$

Question Type : MCQ

Question ID : 8606541354

Option 1 ID : 8606544605

Option 2 ID : 8606544606

Option 3 ID : 8606544603

Option 4 ID : 8606544604

Q.11

Let $f(x) = \int \frac{(2-x^2) \cdot e^x}{(\sqrt{1+x})(1-x)^{3/2}} dx$. If $f(0) = 0$, then $f\left(\frac{1}{2}\right)$ is equal to :

Options

1. $\sqrt{2e} - 1$
2. $\sqrt{2e} + 1$
3. $\sqrt{3e} + 1$
4. $\sqrt{3e} - 1$

Question Type : MCQ

Question ID : 8606541368

Option 1 ID : 8606544662

Option 2 ID : 8606544661

Option 3 ID : 8606544659

Option 4 ID : 8606544660

Q.12

A rectangle is formed by the lines $x=0$, $y=0$, $x=3$ and $y=4$. Let the line L be perpendicular to $3x+y+6=0$ and divide the area of the rectangle into two equal parts. Then the distance of the point

$\left(\frac{1}{2}, -5\right)$ from the line L is equal to :

Options

1. $\sqrt{10}$
2. $2\sqrt{10}$
3. $3\sqrt{10}$
4. $2\sqrt{5}$

Question Type : MCQ

Question ID : 8606541359

Option 1 ID : 8606544625

Option 2 ID : 8606544626

Option 3 ID : 8606544624

Option 4 ID : 8606544623

Q.13 Among the statements :

I: If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{3}{2}$, and

II: If $\begin{vmatrix} x^2 + x & x + 1 & x - 2 \\ 2x^2 + 3x - 1 & 3x & 3x - 3 \\ x^2 + 2x + 3 & 2x - 1 & 2x - 1 \end{vmatrix} = px + q$, then $p^2 = 196q^2$,

Options

1. only II is true
2. only I is true
3. both are false
4. both are true

Question Type : **MCQ**

Question ID : **8606541355**

Option 1 ID : **8606544610**

Option 2 ID : **8606544609**

Option 3 ID : **8606544608**

Option 4 ID : **8606544607**

Q.14 Let $A = \{-2, -1, 0, 1, 2, 3, 4\}$. Let R be a relation on A defined by xRy if and only if $2x + y \leq 2$. Let l be the number of elements in R . Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations respectively. Then $l + m + n$ is equal to :

Options

1. 32
2. 35
3. 34
4. 33

Question Type : **MCQ**

Question ID : **8606541351**

Option 1 ID : **8606544592**

Option 2 ID : **8606544591**

Option 3 ID : **8606544594**

Option 4 ID : **8606544593**

Q.15 A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in :

Options

1. 42 days
2. 30 days
3. 24 days
4. 36 days

Question Type : **MCQ**

Question ID : **8606541353**

Option 1 ID : **8606544602**

Option 2 ID : **8606544600**

Option 3 ID : **8606544599**

Option 4 ID : **8606544601**

Q.16 Let the direction cosines of two lines satisfy the equations : $4l + m - n = 0$ and $2mn + 10nl + 3lm = 0$. Then the cosine of the acute angle between these lines is :

Options

1. $\frac{20}{3\sqrt{38}}$
2. $\frac{10}{3\sqrt{38}}$
3. $\frac{10}{7\sqrt{38}}$
4. $\frac{10}{\sqrt{38}}$

Question Type : **MCQ**

Question ID : **8606541366**

Option 1 ID : **8606544651**

Option 2 ID : **8606544654**

Option 3 ID : **8606544652**

Option 4 ID : **8606544653**

Q.17

The value of $\frac{{}^{100}C_{50}}{51} + \frac{{}^{100}C_{51}}{52} + \dots + \frac{{}^{100}C_{100}}{101}$ is :

Options

1. $\frac{2^{101}}{101}$

2. $\frac{2^{101}}{100}$

3. $\frac{2^{100}}{100}$

4. $\frac{2^{100}}{101}$

Question Type : MCQ

Question ID : 8606541357

Option 1 ID : 8606544618

Option 2 ID : 8606544617

Option 3 ID : 8606544615

Option 4 ID : 8606544616

Q.18

Let α and β respectively be the maximum and the minimum values of the function

$$f(\theta) = 4 \left(\sin^4 \left(\frac{7\pi}{2} - \theta \right) + \sin^4 (11\pi + \theta) \right) - 2 \left(\sin^6 \left(\frac{3\pi}{2} - \theta \right) + \sin^6 (9\pi - \theta) \right), \theta \in \mathbf{R}.$$

Then $\alpha + 2\beta$ is equal to :

Options

1. 4

2. 3

3. 6

4. 5

Question Type : MCQ

Question ID : 8606541363

Option 1 ID : 8606544641

Option 2 ID : 8606544639

Option 3 ID : 8606544640

Option 4 ID : 8606544642

Q.19

Number of solutions of $\sqrt{3} \cos 2\theta + 8 \cos \theta + 3\sqrt{3} = 0$, $\theta \in [-3\pi, 2\pi]$ is :

Options

1. 4
2. 5
3. 3
4. 0

Question Type : MCQ

Question ID : 8606541361

Option 1 ID : 8606544632

Option 2 ID : 8606544631

Option 3 ID : 8606544633

Option 4 ID : 8606544634

Q.20

If α and β ($\alpha < \beta$) are the roots of the equation

$$(-2 + \sqrt{3})(\sqrt{x} - 3) + (x - 6\sqrt{x}) + (9 - 2\sqrt{3}) = 0, x \geq 0, \text{ then } \sqrt{\frac{\beta}{\alpha}} + \sqrt{\alpha\beta} \text{ is equal to :}$$

Options

1. 11
2. 9
3. 10
4. 8

Question Type : MCQ

Question ID : 8606541352

Option 1 ID : 8606544598

Option 2 ID : 8606544596

Option 3 ID : 8606544597

Option 4 ID : 8606544595

Section : Mathematics Section B

Q.21

Let f be a twice differentiable non-negative function such that $(f(x))^2 = 25 + \int_0^x ((f(t))^2 + (f'(t))^2) dt$.

Then the mean of $f(\log_e(1)), f(\log_e(2)), \dots, f(\log_e(625))$ is equal to _____.

Question Type : SA

Question ID : 8606541374

Q.22

Let $|A| = 6$, where A is a 3×3 matrix. If $|\text{adj}(3\text{adj}(A^2 \cdot \text{adj}(2A)))| = 2^m \cdot 3^n$, $m, n \in \mathbf{N}$, then $m + n$ is equal to _____.

Question Type : SA

Question ID : 8606541371

Q.23

The number of 4-letter words, with or without meaning, which can be formed using the letters PQRPRSTUVP, is _____.

Question Type : SA

Question ID : 8606541372

Q.24

Let the area of the region bounded by the curve $y = \max \{ \sin x, \cos x \}$, lines $x = 0$, $x = \frac{3\pi}{2}$, and the x -axis be A . Then, $A + A^2$ is equal to _____.

Question Type : **SA**

Question ID : **8606541375**

Q.25

From the first 100 natural numbers, two numbers first a and then b are selected randomly without replacement. If the probability that $a - b \geq 10$ is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

Question Type : **SA**

Question ID : **8606541373**

Section : Physics Section A

Q.26

Consider light travelling from a medium A to medium B separated by a plane interface. If the light undergoes total internal reflection during its travel from medium A to B and the speed of light in media A and B are 2.4×10^8 m/s and 2.7×10^8 m/s, respectively, then the value of critical angle is :

Options

1. $\sin^{-1}\left(\frac{9}{8}\right)$
2. $\cot^{-1}\left(\frac{3}{\sqrt{13}}\right)$
3. $\cos^{-1}\left(\frac{8}{9}\right)$
4. $\tan^{-1}\left(\frac{8}{\sqrt{17}}\right)$

Question Type : **MCQ**

Question ID : **8606541392**

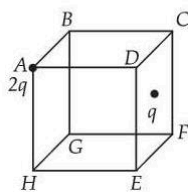
Option 1 ID : **8606544740**

Option 2 ID : **8606544743**

Option 3 ID : **8606544742**

Option 4 ID : **8606544741**

- Q.27** Two point charges $2q$ and q are placed at vertex A and centre of face $CDEF$ of the cube as shown in figure. The electric flux passing through the cube is :



Options

1. $\frac{q}{\epsilon_0}$
2. $\frac{3q}{\epsilon_0}$
3. $\frac{3q}{2\epsilon_0}$
4. $\frac{3q}{4\epsilon_0}$

Question Type : MCQ

Question ID : 8606541389

Option 1 ID : 8606544728

Option 2 ID : 8606544729

Option 3 ID : 8606544730

Option 4 ID : 8606544731

- Q.28** In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg, moving in opposite directions with speeds of 10 m/s and 30 m/s, respectively, strike each other and stick together. The rise in temperature (in $^{\circ}\text{C}$), if all the heat produced during the collision is retained by these spheres, is :
(specific heat of sphere material 31 cal/kg. $^{\circ}\text{C}$ and 1 cal = 4.2 J)

Options

1. 1.15
2. 1.95
3. 1.75
4. 1.44

Question Type : MCQ

Question ID : 8606541383

Option 1 ID : 8606544704

Option 2 ID : 8606544707

Option 3 ID : 8606544706

Option 4 ID : 8606544705

Q.29 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Consider a ferromagnetic material :

Assertion (A) : The individual atoms in a ferromagnetic material possess a magnetic dipole moment and interact with one another in such a way that they spontaneously align themselves forming domains.

Reason (R) : At high enough temperature, the domain structure of ferromagnetic material disintegrates. Thus, magnetization will disappear at high enough temperature known as Curie temperature.

In the light of the above statements, choose the **correct answer** from the options given below :

Options 1.

Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

2. **(A)** is true but **(R)** is false

3.

Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

4. **(A)** is false but **(R)** is true

Question Type : **MCQ**

Question ID : **8606541387**

Option 1 ID : **8606544720**

Option 2 ID : **8606544722**

Option 3 ID : **8606544721**

Option 4 ID : **8606544723**

Q.30 Match List - I with List - II.

List - I
Relation

A. $\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \oint \vec{B} \cdot d\vec{a}$

B. $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt} \right)$

C. $\oint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int_V \rho dv$

D. $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

List - II
Law

I. Ampere's circuital law

II. Faraday's laws of electromagnetic induction

III. Ampere - Maxwell law

IV. Gauss's law of electrostatics

Choose the **correct answer** from the options given below :

Options

1. A-II, B-III, C-I, D-IV

2. A-II, B-III, C-IV, D-I

3. A-I, B-IV, C-III, D-II

4. A-IV, B-I, C-II, D-III

Question Type : **MCQ**

Question ID : **8606541388**

Option 1 ID : **8606544724**

Option 2 ID : **8606544725**

Option 3 ID : **8606544726**

Option 4 ID : **8606544727**

Q.31 In a screw gauge, the zero of the circular scale lies 3 divisions above the horizontal pitch line when their metallic studs are brought in contact. Using this instrument thickness of a sheet is measured. If pitch scale reading is 1 mm and the circular scale reading is 51 then the correct thickness of the sheet is _____ mm.

[Assume least count is 0.01 mm]

Options

1. 1.50
2. 1.51
3. 1.48
4. 1.54

Question Type : MCQ

Question ID : 8606541384

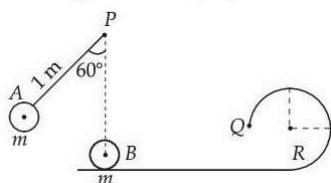
Option 1 ID : 8606544711

Option 2 ID : 8606544710

Option 3 ID : 8606544708

Option 4 ID : 8606544709

Q.32 A small bob A of mass m is attached to a massless rigid rod of length 1 m pivoted at point P and kept at an angle of 60° with vertical as shown in figure. At distance of 1 m below point P, an identical bob B is kept at rest on a smooth horizontal surface that extends to a circular track of radius R as shown in figure. If bob B just manages to complete the circular path of radius R upto a point Q after being hit elastically by bob A, then radius R is _____ m.



Options

1. $\frac{2 - \sqrt{3}}{5}$
2. $\frac{1}{5}$
3. $\frac{3}{5}$
4. $\frac{2 + \sqrt{3}}{5}$

Question Type : MCQ

Question ID : 8606541380

Option 1 ID : 8606544694

Option 2 ID : 8606544693

Option 3 ID : 8606544692

Option 4 ID : 8606544695

Q.33 The de Broglie wavelength of an oxygen molecule at 27°C is $x \times 10^{-12}$ m. The value of x is (take Planck's constant $= 6.63 \times 10^{-34}$ J.s, Boltzmann constant $= 1.38 \times 10^{-23}$ J/K, mass of oxygen molecule $= 5.31 \times 10^{-26}$ kg)

Options

1. 26
2. 24
3. 30
4. 20

Question Type : MCQ

Question ID : 8606541393

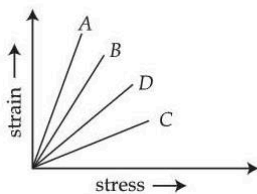
Option 1 ID : 8606544746

Option 2 ID : 8606544745

Option 3 ID : 8606544747

Option 4 ID : 8606544744

Q.34 The strain-stress plot for materials A, B, C and D is shown in the figure. Which material has the largest Young's modulus ?



Options

1. A
2. B
3. D
4. C

Question Type : MCQ

Question ID : 8606541382

Option 1 ID : 8606544700

Option 2 ID : 8606544701

Option 3 ID : 8606544703

Option 4 ID : 8606544702

Q.35 The moment of inertia of a square loop made of four uniform solid cylinders, each having radius R and length L ($R < L$) about an axis passing through the mid points of opposite sides, is (Take the mass of the entire loop as M) :

Options

1. $\frac{3}{4} MR^2 + \frac{1}{6} ML^2$
2. $\frac{3}{8} MR^2 + \frac{1}{6} ML^2$
3. $\frac{3}{4} MR^2 + \frac{7}{12} ML^2$
4. $\frac{3}{8} MR^2 + \frac{7}{12} ML^2$

Question Type : **MCQ**

Question ID : **8606541379**

Option 1 ID : **8606544691**

Option 2 ID : **8606544688**

Option 3 ID : **8606544689**

Option 4 ID : **8606544690**

Q.36 A simple pendulum of string length 30 cm performs 20 oscillations in 10 s. The length of the string required for the pendulum to perform 40 oscillations in the same time duration is _____ cm. [Assume that the mass of the pendulum remains same.]

Options

1. 120
2. 0.75
3. 7.5
4. 15

Question Type : **MCQ**

Question ID : **8606541385**

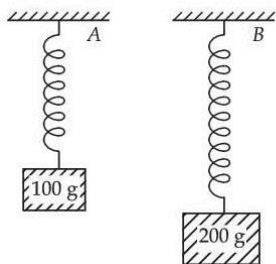
Option 1 ID : **8606544715**

Option 2 ID : **8606544712**

Option 3 ID : **8606544713**

Option 4 ID : **8606544714**

- Q.37** Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B as shown in figure. The energy stored in A is E . The energy stored in B, when spring constants k_A, k_B of A and B, respectively satisfy the relation $4k_A = 3k_B$, is :



Options

1. $\frac{4}{3}E$
2. $2E$
3. $3E$
4. $4E$

Question Type : MCQ

Question ID : 8606541381

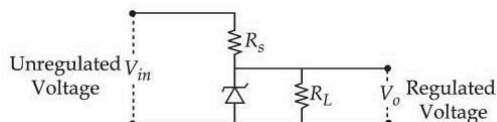
Option 1 ID : 8606544699

Option 2 ID : 8606544698

Option 3 ID : 8606544697

Option 4 ID : 8606544696

- Q.38** The following diagram shows a Zener diode as a voltage regulator. The Zener diode is rated at $V_z = 5$ V and the desired current in load is 5 mA. The unregulated voltage source can supply upto 25 V. Considering the Zener diode can withstand four times of the load current, the value of resistor R_s (shown in circuit) should be _____ Ω .



Options

1. 4000
2. 100
3. 1000
4. 10

Question Type : MCQ

Question ID : 8606541395

Option 1 ID : 8606544755

Option 2 ID : 8606544753

Option 3 ID : 8606544752

Option 4 ID : 8606544754

Q.39 Four persons measure the length of a rod as 20.00 cm, 19.75 cm, 17.01 cm and 18.25 cm. The relative error in the measurement of average length of the rod is :

Options

1. 0.24
2. 0.18
3. 0.06
4. 0.08

Question Type : MCQ

Question ID : 8606541376

Option 1 ID : 8606544679

Option 2 ID : 8606544677

Option 3 ID : 8606544678

Option 4 ID : 8606544676

Q.40 A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produce dispersion without deviation. The angle of second prism is _____.

Options

1. 4.5°
2. 6°
3. 4°
4. 5°

Question Type : MCQ

Question ID : 8606541391

Option 1 ID : 8606544739

Option 2 ID : 8606544738

Option 3 ID : 8606544736

Option 4 ID : 8606544737

- Q.41** Two small balls with masses m and $2m$ are attached to both ends of a rigid rod of length d and negligible mass. If angular momentum of this system is L about an axis (A) passing through its centre of mass and perpendicular to the rod then angular velocity of the system about A is :

Options

1. $\frac{2L}{md^2}$
2. $\frac{2L}{5md^2}$
3. $\frac{4}{3} \frac{L}{md^2}$
4. $\frac{3}{2} \frac{L}{md^2}$

Question Type : **MCQ**

Question ID : **8606541378**

Option 1 ID : **8606544685**

Option 2 ID : **8606544684**

Option 3 ID : **8606544687**

Option 4 ID : **8606544686**

- Q.42** A 20 m long uniform copper wire held horizontally is allowed to fall under the gravity ($g = 10 \text{ m/s}^2$) through a uniform horizontal magnetic field of 0.5 Gauss perpendicular to the length of the wire. The induced EMF across the wire when it travels a vertical distance of 200 m is _____ mV.

Options

1. $0.2\sqrt{10}$
2. $20\sqrt{10}$
3. $200\sqrt{10}$
4. $2\sqrt{10}$

Question Type : **MCQ**

Question ID : **8606541386**

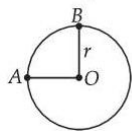
Option 1 ID : **8606544719**

Option 2 ID : **8606544717**

Option 3 ID : **8606544716**

Option 4 ID : **8606544718**

- Q.43** A wire of uniform resistance $\lambda \Omega/\text{m}$ is bent into a circle of radius r and another piece of wire with length $2r$ is connected between points A and B (AOB) as shown in figure. The equivalent resistance between points A and B is _____ Ω .



Options

1. $\frac{3\pi\lambda r}{8}$
2. $(\pi+1)2r\lambda$
3. $2\pi\lambda r$
4. $\frac{6\pi\lambda r}{3\pi + 16}$

Question Type : MCQ

Question ID : 8606541390

Option 1 ID : 8606544732

Option 2 ID : 8606544735

Option 3 ID : 8606544734

Option 4 ID : 8606544733

- Q.44** In hydrogen atom spectrum, ($R \rightarrow$ Rydberg's constant)

- A. the maximum wavelength of the radiation of Lyman series is $\frac{4}{3R}$
- B. the Balmer series lies in the visible region of the spectrum
- C. the minimum wavelength of the radiation of Paschen series is $\frac{9}{R}$
- D. the minimum wavelength of Lyman series is $\frac{5}{4R}$

Choose the **correct** answer from the options given below :

Options

1. A, B and D Only
2. A, B and C Only
3. B, D Only
4. A, B Only

Question Type : MCQ

Question ID : 8606541394

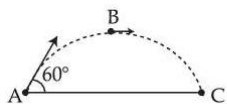
Option 1 ID : 8606544748

Option 2 ID : 8606544750

Option 3 ID : 8606544751

Option 4 ID : 8606544749

- Q.45** An object is projected with kinetic energy K from a point A at an angle 60° with the horizontal. The ratio of the difference in kinetic energies at points B and C to that at point A (see figure), in the absence of air friction is :



- Options**
1. 1 : 2
 2. 3 : 4
 3. 1 : 4
 4. 2 : 3

Question Type : **MCQ**

Question ID : **8606541377**

Option 1 ID : **8606544680**

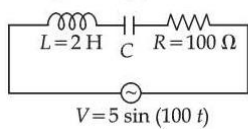
Option 2 ID : **8606544682**

Option 3 ID : **8606544683**

Option 4 ID : **8606544681**

Section : Physics Section B

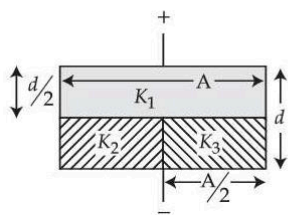
- Q.46** Using a variable frequency a.c. voltage source the maximum current measured in the given LCR circuit is 50 mA for $V = 5 \sin(100t)$. The values of L and R are shown in the figure. The capacitance of the capacitor (C) used is _____ μF .



Question Type : **SA**

Question ID : **8606541396**

- Q.47** The space between the plates of a parallel plate capacitor of capacitance C (without any dielectric) is now filled with three dielectric slabs of dielectric constants $K_1 = 2$, $K_2 = 3$ and $K_3 = 5$ (as shown in figure). If new capacitance is $\frac{n}{3}C$ then the value of n is _____.



Question Type : **SA**

Question ID : **8606541400**

- Q.48** The equation of the electric field of an electromagnetic wave propagating through free space is given by : $E = \sqrt{377} \sin(6.27 \times 10^3 t - 2.09 \times 10^{-5} x)$ N/C

The average power of the electromagnetic wave is $\left(\frac{1}{\alpha}\right) \text{ W/m}^2$. The value of α is _____

(Take $\sqrt{\frac{\mu_0}{\epsilon_0}} = 377$ in SI units)

Question Type : **SA**

Question ID : **8606541398**

Q.49 In two separate Young's double-slit experimental set-ups and two monochromatic light sources of different wavelengths are used to get fringes of equal width. The ratios of the slits separations and that of the wavelengths of light used are 2 : 1 and 1 : 2 respectively. The corresponding ratio of the distances between the slits and the respective screens (D_1/D_2) is _____.

Question Type : SA

Question ID : 8606541397

Q.50 A simple pendulum made of mass 10 g and a metallic wire of length 10 cm is suspended vertically in a uniform magnetic field of 2 T. The magnetic field direction is perpendicular to the plane of oscillations of the pendulum. If the pendulum is released from an angle of 60° with vertical, then maximum induced EMF between the point of suspension and point of oscillation is _____ mV. (Take $g = 10 \text{ m/s}^2$)

Question Type : SA

Question ID : 8606541399

Section : **Chemistry Section A**

Q.51 The correct trend in the first ionization enthalpies of the elements in the 3rd period of periodic table is :

Options

1. $\text{Si} < \text{S} < \text{Al} < \text{P} < \text{Cl}$
2. $\text{Al} < \text{S} < \text{P} < \text{Si} < \text{Cl}$
3. $\text{S} < \text{Si} < \text{Al} < \text{P} < \text{Cl}$
4. $\text{Al} < \text{Si} < \text{S} < \text{P} < \text{Cl}$

Question Type : MCQ

Question ID : 8606541408

Option 1 ID : 8606544789

Option 2 ID : 8606544792

Option 3 ID : 8606544791

Option 4 ID : 8606544790

Q.52 Given,
(A) $n=5, m_l = -1$

(B) $n=3, l=2, m_l = -1, m_s = +\frac{1}{2}$

The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

Options

1. 8 and 1
2. 2 and 4
3. 26 and 1
4. 4 and 1

Question Type : MCQ

Question ID : 8606541402

Option 1 ID : 8606544767

Option 2 ID : 8606544766

Option 3 ID : 8606544765

Option 4 ID : 8606544768

Q.53 Which of the following statements regarding the energy of the stationary state is **true** in the following one - electron systems ?

Options

1. -2.18×10^{-18} J for third orbit of Li^{2+} ion
2. $+2.18 \times 10^{-18}$ J for second orbit of He^+ ion
3. $+8.72 \times 10^{-18}$ J for first orbit of He^+ ion
4. -1.09×10^{-18} J for second orbit of H atom.

Question Type : **MCQ**

Question ID : **8606541401**

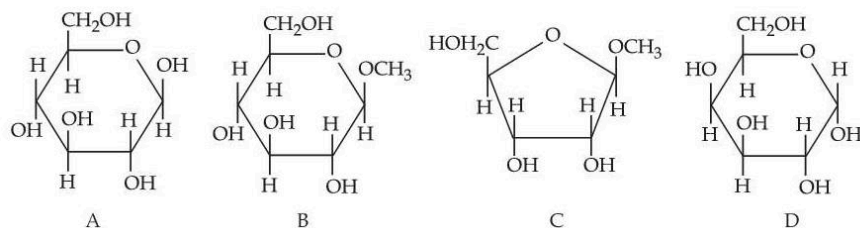
Option 1 ID : **8606544764**

Option 2 ID : **8606544762**

Option 3 ID : **8606544763**

Option 4 ID : **8606544761**

Q.54 From the given following (A to D) cyclic structures, those which will not react with Tollen's reagent are :



Options

1. A and D
2. B and D
3. B and C
4. A and B

Question Type : **MCQ**

Question ID : **8606541420**

Option 1 ID : **8606544838**

Option 2 ID : **8606544839**

Option 3 ID : **8606544840**

Option 4 ID : **8606544837**

Q.55 Given below are two statements :

Statement I : Sublimation is used for the separation and purification of compounds with low melting point.

Statement II : The boiling point of a liquid increases as the external pressure is reduced.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. **Statement I is true but Statement II is false**
2. **Statement I is false but Statement II is true**
3. **Both Statement I and Statement II are true**
4. **Both Statement I and Statement II are false**

Question Type : **MCQ**

Question ID : **8606541413**

Option 1 ID : **8606544811**

Option 2 ID : **8606544812**

Option 3 ID : **8606544809**

Option 4 ID : **8606544810**

Q.56 A cup of water at 5°C (system) is placed in a microwave oven and the oven is turned on for one minute during which the water begins to boil. Which of the following option is **true** ?

Options

1. **$q = +ve, w = 0, \Delta U = -ve$**
2. **$q = +ve, w = -ve, \Delta U = +ve$**
3. **$q = -ve, w = -ve, \Delta U = -ve$**
4. **$q = +ve, w = -ve, \Delta U = -ve$**

Question Type : **MCQ**

Question ID : **8606541404**

Option 1 ID : **8606544774**

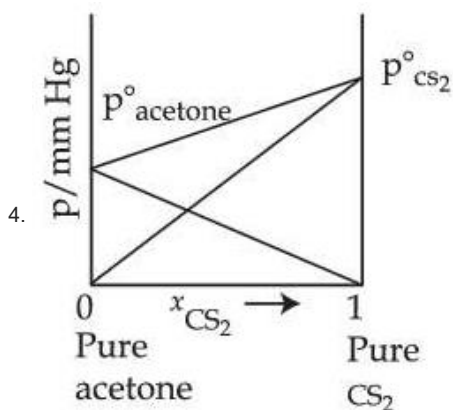
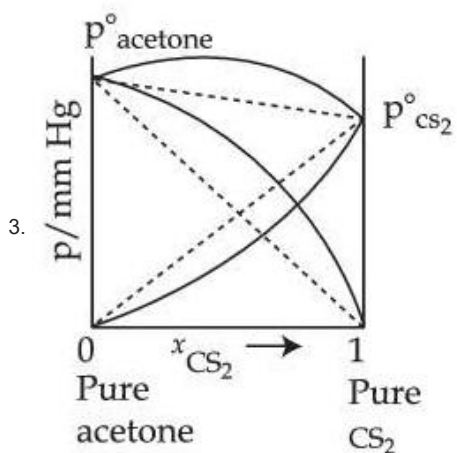
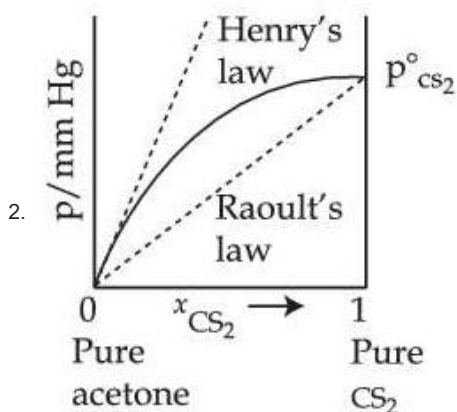
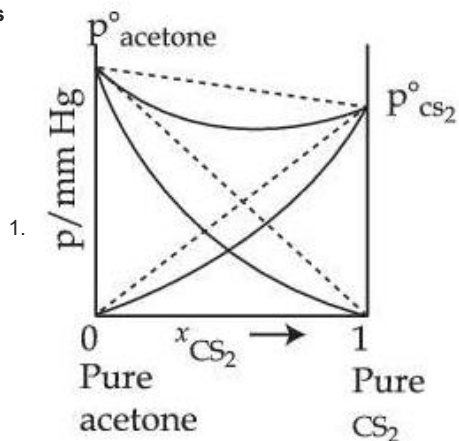
Option 2 ID : **8606544773**

Option 3 ID : **8606544776**

Option 4 ID : **8606544775**

Q.57 Which one of the following graphs accurately represents the plot of partial pressure of CS_2 vs its mole fraction in a mixture of acetone and CS_2 at constant temperature?

Options



Q.58 In the given electrochemical cell, $\text{Ag(s)} | \text{AgCl(s)} | \text{FeCl}_2(\text{aq}), \text{FeCl}_3(\text{aq}) | \text{Pt(s)}$ at 298 K, the cell potential (E_{cell}) will increase when :

- Concentration of Fe^{2+} is increased.
- Concentration of Fe^{3+} is decreased.
- Concentration of Fe^{2+} is decreased.
- Concentration of Fe^{3+} is increased.
- Concentration of Cl^- is increased.

Choose the **correct** answer from the options given below :

Options

- C, D and E Only
- A and B Only
- B Only
- A and E Only

Question Type : **MCQ**

Question ID : 8606541407

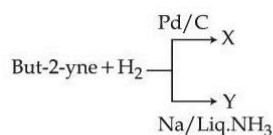
Option 1 ID : 8606544788

Option 2 ID : 8606544785

Option 3 ID : 8606544787

Option 4 ID : 8606544786

Q.59 But-2-yne and hydrogen (one mole each) are separately treated with (i) Pd/C and (ii) Na/liq. NH_3 to give the products X and Y respectively.



Identify the **incorrect** statements.

- X and Y are stereoisomers.
- Dipole moment of X is zero.
- Boiling point of X is higher than Y.
- X and Y react with $\text{O}_3/\text{Zn} + \text{H}_2\text{O}$ to give different products.

Choose the **correct** answer from the options given below :

Options

- A and C Only
- B and C Only
- A and B Only
- B and D Only

Question Type : **MCQ**

Question ID : 8606541415

Option 1 ID : 8606544817

Option 2 ID : 8606544819

Option 3 ID : 8606544820

Option 4 ID : 8606544818

Q.60 Match List - I with List - II.

List - I

Functional group (detection)

- A. Unsaturation (Baeyer's test)
- B. Alcoholic group (Ceric ammonium nitrate test)
- C. Aldehyde group (Tollen's reagent)
- D. Phenolic group (FeCl_3 test)

List - II

Change observed during detection

- I. Red colour appears
- II. Silver mirror appears
- III. Violet colour appears
- IV. Discharge of pink colour

Choose the **correct** answer from the options given below :

Options

- 1. A-IV, B-III, C-II, D-I
- 2. A-III, B-IV, C-I, D-II
- 3. A-III, B-IV, C-II, D-I
- 4. A-IV, B-I, C-II, D-III

Question Type : **MCQ**

Question ID : **8606541412**

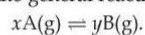
Option 1 ID : **8606544805**

Option 2 ID : **8606544808**

Option 3 ID : **8606544806**

Option 4 ID : **8606544807**

Q.61 Consider the general reaction given below at 400 K



The values of K_p and K_c are studied under the same condition of temperature but variation in x and y .

(i) $K_p = 85.87$ and $K_c = 2.586$ appropriate units

(ii) $K_p = 0.862$ and $K_c = 28.62$ appropriate units

The values of x and y in (i) and (ii) respectively are :

(i) (ii)

Options

- 1. 3, 1 3, 1
- 2. 4, 1 4, 1
- 3. 1, 2 2, 1
- 4. 1, 3 2, 1

Question Type : **MCQ**

Question ID : **8606541406**

Option 1 ID : **8606544783**

Option 2 ID : **8606544784**

Option 3 ID : **8606544781**

Option 4 ID : **8606544782**

Q.62 Given below are two statements :

Statement I : $[\text{CoBr}_4]^{2-}$ ion will absorb light of lower energy than $[\text{CoCl}_4]^{2-}$ ion.

Statement II : In $[\text{CoI}_4]^{2-}$ ion, the energy separation between the two set of d-orbitals is more than $[\text{CoCl}_4]^{2-}$ ion.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. Both **Statement I** and **Statement II** are false
2. Both **Statement I** and **Statement II** are true
3. **Statement I** is false but **Statement II** is true
4. **Statement I** is true but **Statement II** is false

Question Type : **MCQ**

Question ID : **8606541410**

Option 1 ID : **8606544798**

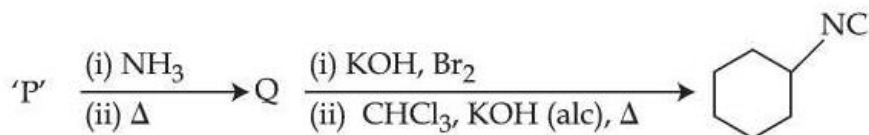
Option 2 ID : **8606544797**

Option 3 ID : **8606544800**

Option 4 ID : **8606544799**

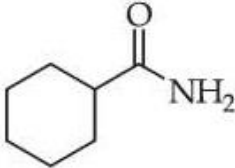
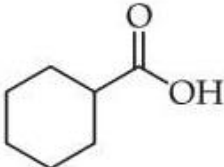
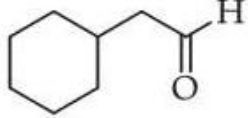
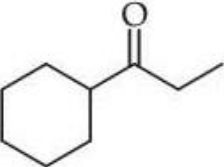
Q.63

Compound 'P' undergoes the following sequence of reactions :



'P' is :

Options

1. 
2. 
3. 
4. 

Question Type : **MCQ**

Question ID : **8606541418**

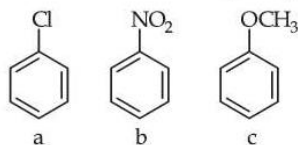
Option 1 ID : **8606544831**

Option 2 ID : **8606544829**

Option 3 ID : **8606544830**

Option 4 ID : **8606544832**

Q.64 Consider the following compounds



Arrange these compounds in the increasing order of reactivity with nitrating mixture.

Options

1. $b < c < a$
2. $c < b < a$
3. $b < a < c$
4. $c < a < b$

Question Type : MCQ

Question ID : 8606541414

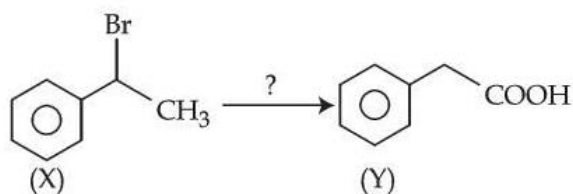
Option 1 ID : 8606544814

Option 2 ID : 8606544813

Option 3 ID : 8606544816

Option 4 ID : 8606544815

Q.65



The correct sequence of reagents for the above conversion of X to Y is :

Options 1.

1. (i) NaOH (aq) (ii) Jones reagent (iii) H_3O^+
2. (i) Jones reagent (ii) NaOEt (iii) Hot KMnO_4/KOH
3. (i) $\text{B}_2\text{H}_6/\text{H}_2\text{O}_2$ (ii) NaOEt (iii) Jones reagent
4. (i) NaOEt (ii) $\text{B}_2\text{H}_6/\text{H}_2\text{O}_2$ (iii) Jones reagent

Question Type : MCQ

Question ID : 8606541416

Option 1 ID : 8606544824

Option 2 ID : 8606544822

Option 3 ID : 8606544821

Option 4 ID : 8606544823

Q.66 'x' is the product which is obtained from propanenitrile and stannous chloride in the presence of hydrochloric acid followed by hydrolysis. 'y' is the product which is obtained from the but-2-ene by the ozonolysis followed by hydrolysis. From the following, which product is not obtained when one mole of 'x' and one mole of 'y' react with each other in the presence of alkali followed by heating?

Options

1. Pent-2-enal
2. 2-Methylpent-2-enal
3. 2-Methylbut-2-enal
4. 3-Methylbut-2-enal

Question Type : MCQ

Question ID : 8606541417

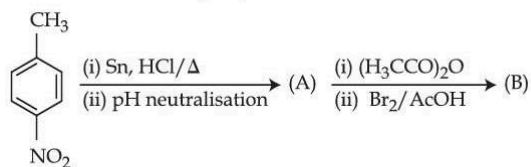
Option 1 ID : 8606544826

Option 2 ID : 8606544827

Option 3 ID : 8606544828

Option 4 ID : 8606544825

Q.67 Consider the following sequence of reactions.



4-Nitrotoluene

Assuming that the reaction proceeds to completion, then 137 mg of 4-nitrotoluene will produce _____ mg of B.

(Given molar mass in g mol^{-1} H : 1, C : 12, N : 14, O : 16, Br : 80)

Options

1. 208
2. 228
3. 301
4. 146

Question Type : MCQ

Question ID : 8606541419

Option 1 ID : 8606544833

Option 2 ID : 8606544834

Option 3 ID : 8606544836

Option 4 ID : 8606544835

- Q.68** The correct statements from the following are :
- A. Ionic radii of trivalent cations of group 13 elements decreases down the group.
 - B. Electronegativity of group 13 elements decreases down the group.
 - C. Among the group 13 elements, Boron has highest first ionisation enthalpy.
 - D. The trichloride and triiodide of group 13 elements are covalent in nature.
- Choose the **correct** answer from the options given below :

Options

- 1. A and C Only
- 2. C and D Only
- 3. B and D Only
- 4. A and D Only

Question Type : **MCQ**

Question ID : **8606541409**

Option 1 ID : **8606544793**

Option 2 ID : **8606544795**

Option 3 ID : **8606544794**

Option 4 ID : **8606544796**

- Q.69** The statements that are **incorrect** about the nickel(II) complex of dimethylglyoxime are :
- A. It is red in colour.
 - B. It has a high solubility in water at pH=9.
 - C. The Ni ion has two unpaired d-electrons.
 - D. The N – Ni – N bond angle is almost close to 90°.
 - E. The complex contains four five-membered metallacycles (metal containing rings).
- Choose the **correct** answer from the options given below :

Options

- 1. B, C and E Only
- 2. C and D Only
- 3. C and E Only
- 4. A, D and B Only

Question Type : **MCQ**

Question ID : **8606541411**

Option 1 ID : **8606544801**

Option 2 ID : **8606544803**

Option 3 ID : **8606544804**

Option 4 ID : **8606544802**

Q.70 Identify the molecule (X) with maximum number of lone pairs of electrons (obtained using Lewis dot structure) among HNO_3 , H_2SO_4 , NF_3 and O_3 . Choose the **correct** bond angle made by the central atom of the molecule (X).

Options

1. 107°
2. 120°
3. 102°
4. 116°

Question Type : **MCQ**

Question ID : **8606541403**

Option 1 ID : **8606544771**

Option 2 ID : **8606544772**

Option 3 ID : **8606544770**

Option 4 ID : **8606544769**

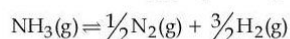
Section : **Chemistry Section B**

Q.71 The crystal field splitting energy of $[\text{Co}(\text{oxalate})_3]^{3-}$ complex is 'n' times that of the $[\text{Cr}(\text{oxalate})_3]^{3-}$ complex. Here 'n' is _____. (Assume $\Delta_o > P$)

Question Type : **SA**

Question ID : **8606541424**

Q.72 For the following gas phase equilibrium reaction at constant temperature,



if the total pressure is $\sqrt{3}$ atm and the pressure equilibrium constant (K_p) is 9 atm, then the degree of dissociation is given as $(x \times 10^{-2})^{-1/2}$. The value of x is _____. (nearest integer)

Question Type : **SA**

Question ID : **8606541422**

Q.73 x mg of pure HCl was used to make an aqueous solution. 25.0 mL of 0.1 M $\text{Ba}(\text{OH})_2$ solution is used when the HCl solution was titrated against it. The numerical value of x is _____ $\times 10^{-1}$. (Nearest integer)

Given : Molar mass of HCl and $\text{Ba}(\text{OH})_2$ are 36.5 and 171.0 g mol^{-1} respectively.

Question Type : **SA**

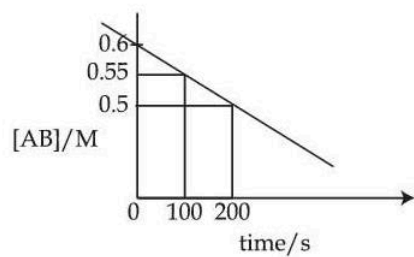
Question ID : **8606541421**

Q.74 Consider all the structural isomers with molecular formula $\text{C}_5\text{H}_{11}\text{Br}$ are separately treated with $\text{KOH}(\text{aq})$ to give respective substitution products, without any rearrangement. The number of products which can exhibit optical isomerism from these is _____.

Question Type : **SA**

Question ID : **8606541425**

Q.75 For the thermal decomposition of reactant $AB(g)$, the following plot is constructed.



The half life of the reaction is ' x ' min.

$x =$ _____ min. (Nearest integer)

Question Type : **SA**

Question ID : **8606541423**